```
============================================================
  MILESTONE 2: Feature Extraction & Modeling
============================================================

[Feature Extraction] Computing rolling statistical features...
   ✅ Feature matrix shape: (17280, 35)
   💾 Feature matrix → outputs/feature_matrix.csv

[Prophet] Modeling 'heart_rate_bpm'...
02:19:03 - cmdstanpy - INFO - Chain [1] start processing
02:19:03 - cmdstanpy - INFO - Chain [1] done processing
   ✅ Prophet anomalies detected: 82 (5.7%)
   📊 Prophet plot → outputs/prophet_heart_rate_bpm.png

[Prophet] Modeling 'steps'...
02:19:04 - cmdstanpy - INFO - Chain [1] start processing
02:19:05 - cmdstanpy - INFO - Chain [1] done processing
   ✅ Prophet anomalies detected: 58 (4.0%)
   📊 Prophet plot → outputs/prophet_steps.png

[Prophet] Modeling 'spo2_pct'...
02:19:06 - cmdstanpy - INFO - Chain [1] start processing
02:19:06 - cmdstanpy - INFO - Chain [1] done processing
   ✅ Prophet anomalies detected: 85 (5.9%)
   📊 Prophet plot → outputs/prophet_spo2_pct.png

[Clustering] Running KMeans (k=4) and DBSCAN...
   ✅ KMeans clusters: [np.int32(0), np.int32(1), np.int32(2), np.int32(3)]
   ✅ DBSCAN outliers: 965
   📊 Cluster plot → outputs/milestone2_clusters.png

✅ Milestone 2 Complete!
```

FOLDERS

INFO

outputs

- anomaly_results.csv
- cleaned_data.csv
- feature_clustered.csv
- feature_matrix.csv
- milestone1_preview....
- milestone2_clusters....
- milestone3_heartrate...
- milestone3_sleep.png
- milestone3_steps_sp...
- milestone3_summary...
- prophet_heart_rate_...
- prophet_spo2_pct.png
- prophet_steps.png
- fitness_data_raw.csv
- milestone1_preproces...
- milestone2_modeling....
- milestone3_anomaly.py
- run_all.py

> OUTLINE

> TIMELINE

feature_matrix.csv

outputs > feature_matrix.csv > data

Col 5: heart_rate_bpm_max

```
timestamp,heart_rate_bpm_mean,heart_rate_bpm_std,heart_rate_bpm_min,heart_rate_bpm_max,heart_rate_bpm_range,heart_rate_bpm_skew,heart_rate_bpm_kurt,heart_rate_bpm_energy,steps_mean,steps_std,steps_min,steps_max,steps_range,steps_
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2024-01-01 03:55:00,58.525000000000006,5.2440659104102 83,51.2,66.4,15.200000000000003,0.3096609844885317,-1.3666468856831562,41404.60999999999,0.0,0.0,0.0,0.0,0.0,0.0,0.0,-3.0,0.0,97.70833333333333,0.7102603022622672,96.6,99.4,2.800
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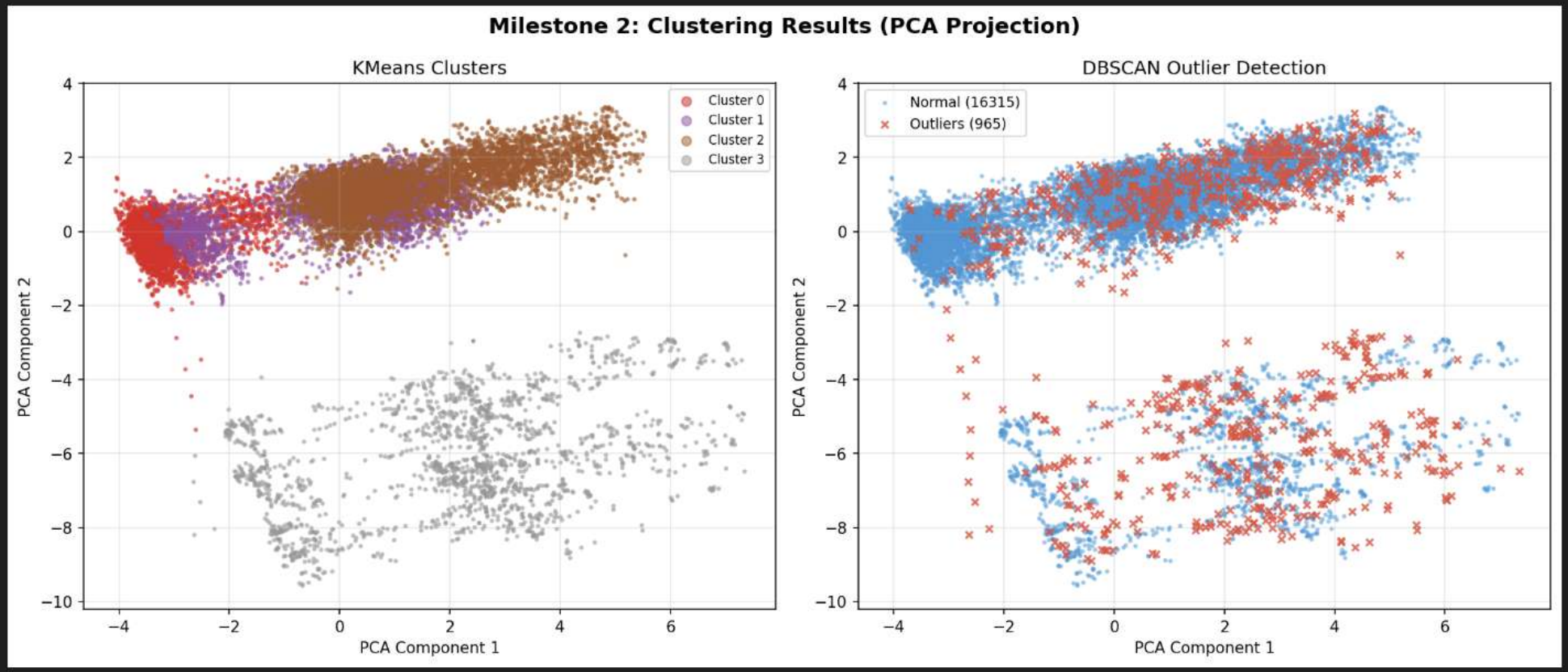
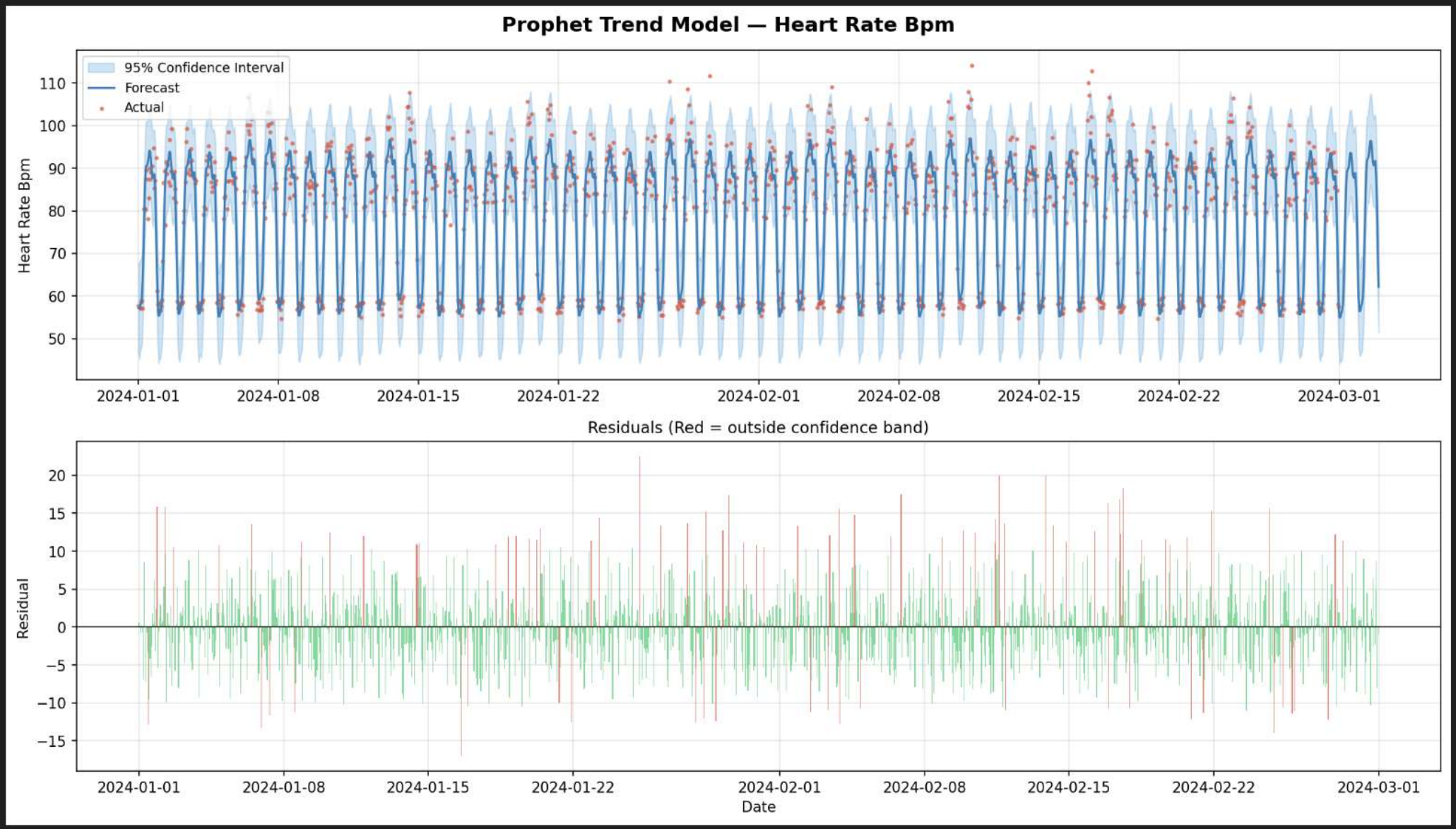
milestone2_clusters.png

outputs > milestone2_clusters.png



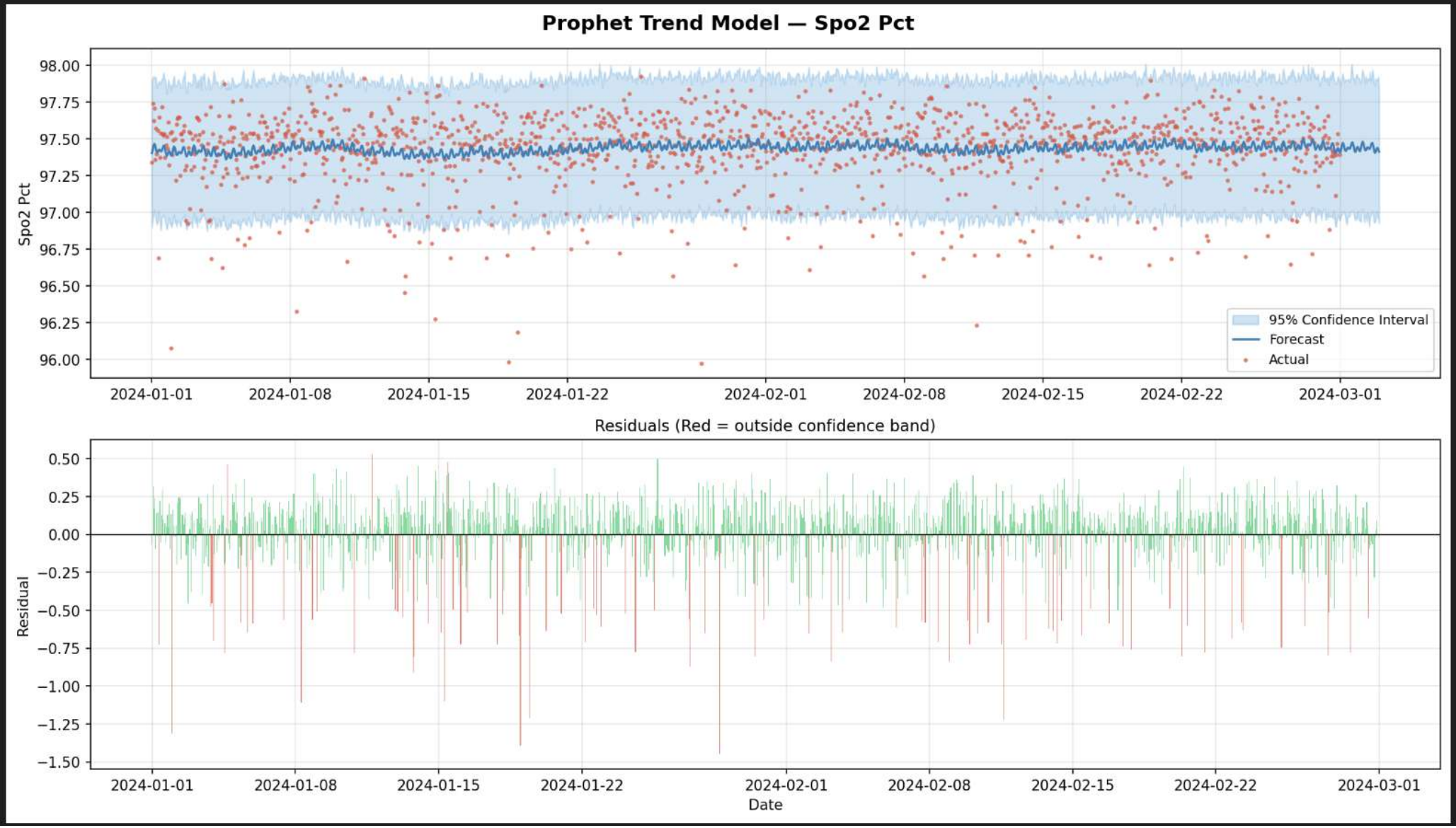
prophet_heart_rate_bpm.png

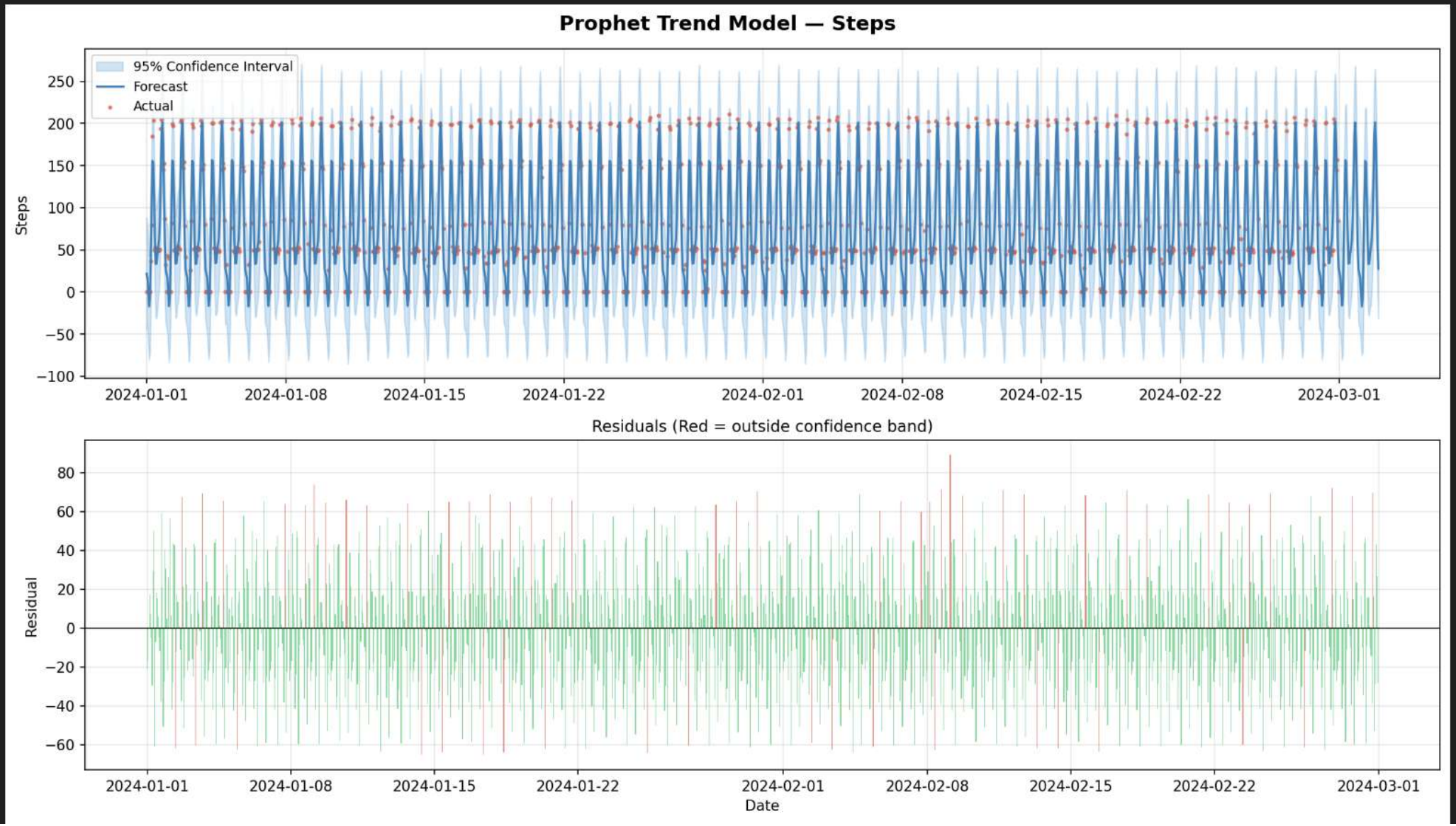
outputs > prophet_heart_rate_bpm.png



prophet_spo2_pct.png

outputs > prophet_spo2_pct.png



prophet_steps.png

outputs > prophet_steps.png